

Import Dependencies of Dutch Firms

Extended Abstract – Preliminary Draft, Please Do Not Circulate

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1. Introduction

Global value chains have made firms increasingly interconnected across countries, improving efficiency but also creating new vulnerabilities to external shocks. While international trade has enhanced specialization and productivity, it has also raised concerns about exposure to disruptions in international supply chains. Recent geopolitical tensions, disruptions during the covid-19 pandemic and increased attention to critical raw materials have intensified policy debates about supply chain resilience and strategic dependencies.

These concerns are particularly relevant for highly open economies such as the Netherlands, where imports play a crucial role in firms' production processes. Access to international suppliers enables firms to benefit from specialization and cost advantages, but it may also create vulnerabilities when production relies on a limited number of foreign sources. In such cases, disruptions in global value chains can directly affect firms' ability to obtain inputs and maintain production.

Identifying import dependencies at the firm level is important for informing policies aimed at strengthening supply chain resilience. In recent years, policymakers have become increasingly concerned about the vulnerability of production systems to disruptions arising from geopolitical tensions, trade restrictions, or global crises. By identifying which firms rely on highly concentrated sources for products that are difficult to substitute internationally or domestically, the analysis provides insights into where potential vulnerabilities are located within the economy. Such information can help policymakers better assess supply chain risks, design targeted resilience strategies, and evaluate whether certain sectors or products warrant closer monitoring or policy support in the context of increasing uncertainty in global trade.

Assessing import dependence requires going beyond simple measures of sourcing diversification. While insufficient diversification of sourcing may expose firms to heightened risks in the event of supply chain disruptions, focusing solely on the number of sourcing countries provides an incomplete picture. In particular, such measures do not account for whether imported products can be substituted through alternative foreign suppliers or through domestic production.

This study therefore develops a broader approach to identifying import dependencies at the firm level. Specifically, we identify and characterize import dependencies among Dutch firms by combining firm-level trade data with product- and country-level indicators of supply concentration and substitutability. In particular, the analysis seeks to determine whether firms rely on highly concentrated sources for specific products and whether such dependence persists when taking into account the availability of alternative suppliers in global markets and the potential for substitution through domestic production.

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Import dependence is assessed using a multidimensional framework that captures several sources of potential vulnerability. Specifically, import dependence is evaluated along several dimensions: the concentration of a firm's imports across sourcing countries, the concentration of global exports of the imported products, the degree to which these products can be substituted by domestic production, and the persistence of import concentration over time. By combining these dimensions, the analysis identifies firms whose sourcing patterns may expose them to heightened risks in the event of international supply disruptions.

This study contributes to the growing literature on supply chain resilience and international trade dependencies by providing a firm-level assessment of import dependence in a highly open economy. While existing research (European Commission, 2023; Lefebvre & Wibaux, 2024; Vicard & Wibaux, 2023) has largely examined supply chain vulnerabilities at the country or sector level, this study shifts the focus to the firm level, allowing for the identification of heterogeneity in sourcing patterns across firms within the same industries. By combining detailed firm-level import data with product- and country-level indicators of global supply concentration and domestic substitutability, the analysis develops a multidimensional framework to identify import dependencies and examine how such dependencies relate to firm characteristics, including firm size, productivity, and firms' positions in global value chains. Empirically, the paper provides new evidence on the extent and characteristics of import dependencies among Dutch firms and investigates which types of firms are more likely to exhibit such dependencies.

Beyond identifying import-dependent firms, the study also examines their position within the Dutch firm network to assess the potential systemic implications of supply disruptions. In particular, the analysis investigates whether disruptions affecting import-dependent firms could propagate through production linkages and generate contagion effects across the domestic production network. By evaluating the network position of dependent firms via several centrality measures, the study provides insights into the extent to which firm-level import dependencies may translate into broader systemic risks for the Dutch economy.

2. Data and Methodology

This study uses detailed firm-level trade data for Dutch firms that record imports at the firm–product–country level. The data are obtained from Statistics Netherlands (CBS). These data allow us to observe which firms import which products, from which countries, and in what value. The firm-level import data are combined with international trade statistics from the BACI International Trade Database developed by CEPII to capture the global export structure of products.

In addition, firm-level characteristics, such as firm size, productivity and indicators related to firms' participation in global value chains, are incorporated using firm-level microdata from CBS. These data allow us to analyze how import dependence relates to firm heterogeneity. Finally, the dataset is linked to inter-firm transaction data from CBS that map domestic buyer–supplier relationships within the Netherlands, enabling the analysis of the position of import-dependent firms within the Dutch production network.

Import dependence is identified using a multidimensional framework that captures several potential sources of vulnerability in firms' sourcing structures. The first dimension measures the concentration of imports across sourcing countries at the firm–product level. For each firm f , product i , and year t , import concentration is measured using the Herfindahl–Hirschman Index (HHI):

$$HHI_{f,i,t} = \sum_{c \in C} \left[\frac{M_{f,c,i,t}}{M_{f,i,t}} \right]^2, \forall f \in F, \forall i \in I, \forall t \in T$$

Where $M_{f,c,i,t}$ denotes the value of imports of product i by firm f from country c in year t , and $M_{f,i,t}$ represents the total imports of product i by firm f in year t . Higher values of this index indicate that imports are concentrated among a small number of supplier countries, suggesting a higher degree of sourcing dependence.

The second dimension evaluates the concentration of global supply for each product, capturing the availability of alternative foreign suppliers. Global export concentration is calculated using a similar Herfindahl index:

$$HHI_{i,t}^{world} = \sum_{e \in E} \left[\frac{X_{e,i,t}}{X_{i,t}} \right]^2, \forall i \in I, \forall t \in T$$

where $X_{e,i,t}$ denotes exports of product i by exporting country e in year t and $X_{i,t}$ denotes total world exports of product i in year t . A high value of this index indicates that global supply is dominated by a limited number of exporting countries, implying limited international substitutability.

The third dimension assesses whether imported products could potentially be substituted by domestic production. This is measured by comparing Dutch imports and exports of each product:

$$S_{i,t}^{NLD} = \frac{M_{i,t}}{X_{i,t}}, \forall i \in I, \forall t \in T$$

here $M_{i,t}$ represents total imports of product i into the Netherlands and $X_{i,t}$ denotes Dutch exports of the same product in year t . Values greater than one suggest that imports exceed domestic export capacity, indicating limited potential for domestic substitution.

Finally, the framework considers the persistence of import concentration over time to distinguish structural dependence from temporary fluctuations in sourcing patterns. By combining firm-level import concentration, global export concentration, domestic substitutability, and persistence over time, the analysis identifies firms whose sourcing structures may expose them to heightened vulnerability in the event of international supply disruptions.

After identifying import-dependent firms, the study examines how import dependence relates to firm characteristics such as firm size, productivity, and firms' positions in global value chains. In particular, the analysis investigates whether reducing dependence on a specific supplier can be interpreted as a form of insurance against supply chain disruptions, and which types of firms are more likely to adopt such diversification strategies. This allows us to assess whether certain firms, such as larger, more productive firms or firms positioned more upstream or downstream in global value chains, are better able or more inclined to mitigate supply risks through more diversified sourcing structures.

Furthermore, the analysis evaluates the position of import-dependent firms within the Dutch production network to assess whether supply disruptions affecting these firms could

propagate through inter-firm linkages and generate broader contagion or systemic risks within the domestic economy.

3. Results

Figures 1 and 2 provide an overview of the importing behavior of Dutch firms. Figure 1 shows that the number of importing firms increased steadily between 2009 and 2024, rising from around 45,000 firms to more than 60,000 firms, with a temporary decline around 2020. Over the same period, the average number of imported products per firm remained relatively stable, fluctuating between roughly 15 and 18 products. While this average suggests a moderate level of diversification, Figure 2 reveals substantial heterogeneity across firms. The distribution of imported products per firm is highly skewed, with the majority of firms importing only a small number of products, while a relatively small group of firms imports a very large number of products.

Figure 1. Trends in the Number of Importing Firms and the Average Number of Imported Products per Firm in the Netherlands (2009–2024)

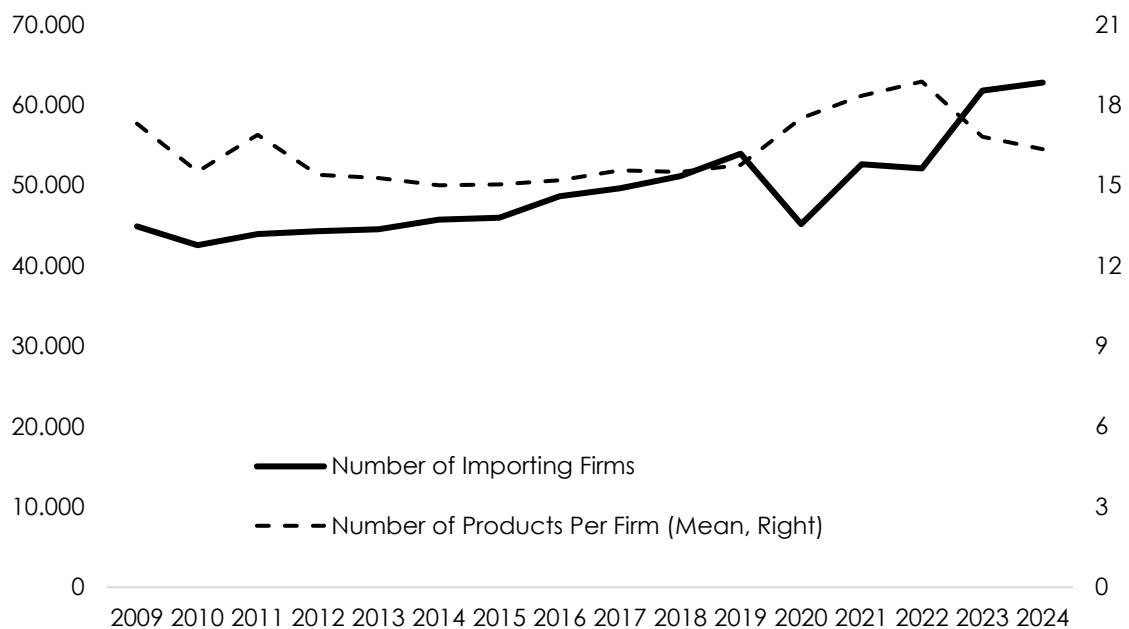


Figure 2. Distribution of the Number of Imported Products per Firm (2009–2024)

Figures 3 and 4 describe firms' sourcing patterns across import partners. Figure 3 shows that the average number of products imported from each partner country increased slightly over time, peaking around 2021 before declining modestly in recent years. Figure 4 shows that most firm–product imports are sourced from a single country, consistently accounting for around 70 percent of observations, while imports from two countries represent a much smaller share and sourcing from three or more countries remains rare. Together, these figures suggest that firms tend to rely on a limited number of supplier countries for each product, indicating a relatively concentrated sourcing structure.

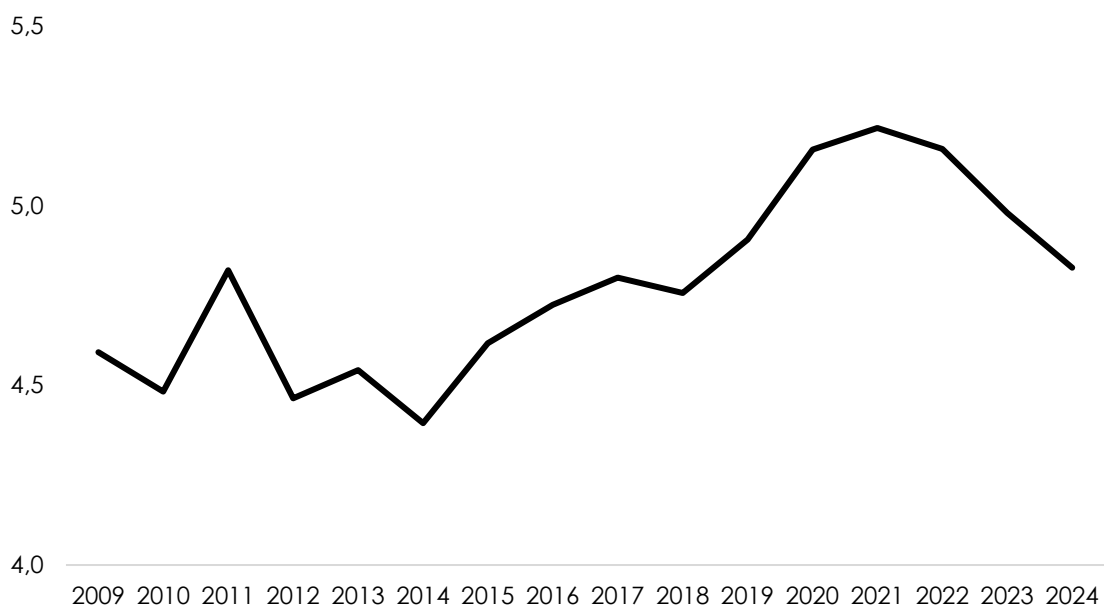
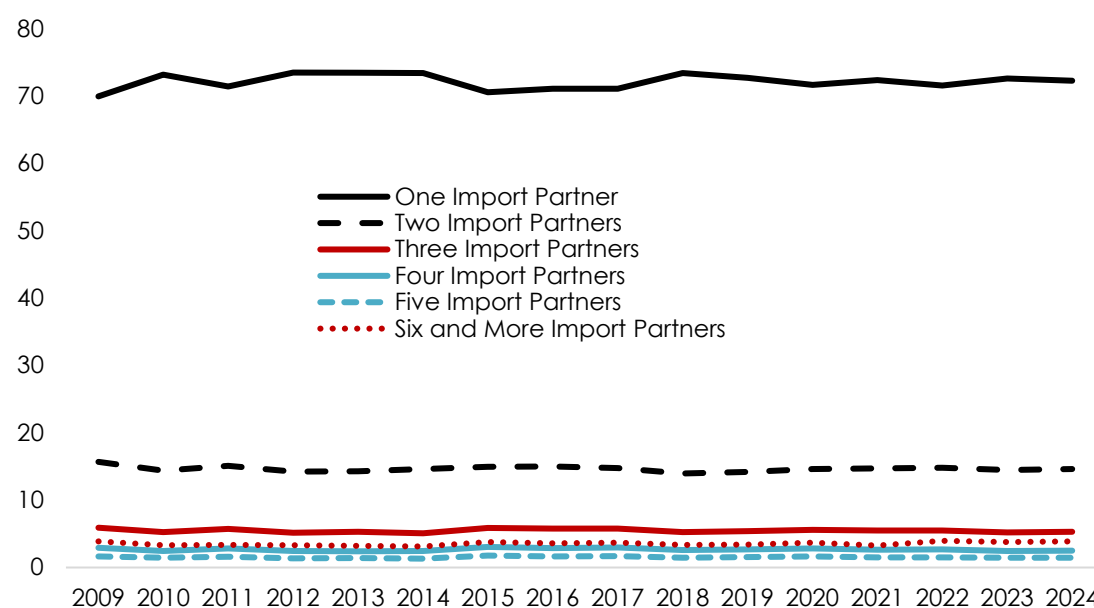
Figure 3. Average Number of Imported Products per Import Partner, 2009–2024

Figure 4. Share of Firm–Product Imports by Number of Import Partner Countries, 2009–2024

Figures 5 and 6 provide further evidence on the concentration of firms' sourcing across countries. Figure 5 shows that, on average, a single import partner accounts for a relatively large share of a firm's total imports, increasing from around 52 percent in 2009 to about 60 percent in 2024, with a temporary decline around 2020. Figure 6 highlights the distribution of this measure across firm–country pairs and shows a highly skewed pattern: many observations are clustered at very low shares, while a substantial number are concentrated near very high shares. This indicates that although firms often import small amounts from multiple countries, a large portion of their total imports typically originates from a dominant supplier country, suggesting a high degree of concentration in firms' sourcing structures.

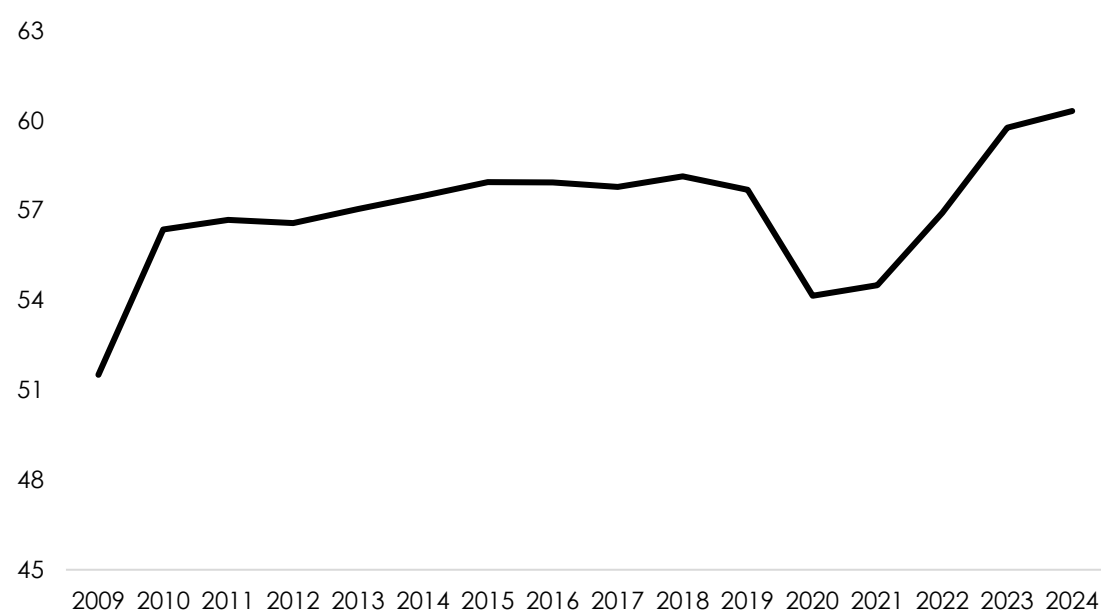
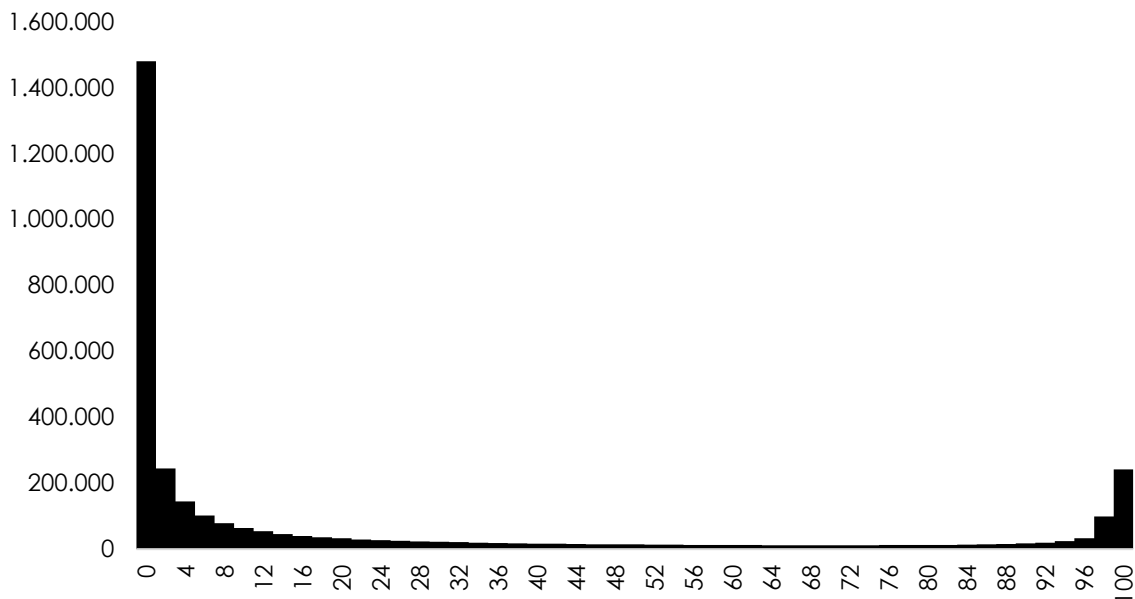
Figure 5. Mean Share of Firms' Imports per Country in Total Firm Imports (%), 2009–2024

Figure 6. Distribution of the Share of Firms' Imports per Import Country in Total Firm Imports

Figures 7 and 8 illustrate the degree of concentration in firms' sourcing structures. Figure 7 shows the distribution of the share of total firm imports that originate from the largest supplier country. The distribution is heavily skewed toward high values, with a large number of firms sourcing the vast majority of their imports from a single country. Figure 8 presents a similar pattern when looking at the share of products imported from the dominant supplier country. A significant number of firms rely on a single partner country for most of the products they import. Together, these figures highlight the strong concentration of import relationships at the firm level, suggesting that many firms depend heavily on a dominant foreign supplier for both the value and the range of their imported products.

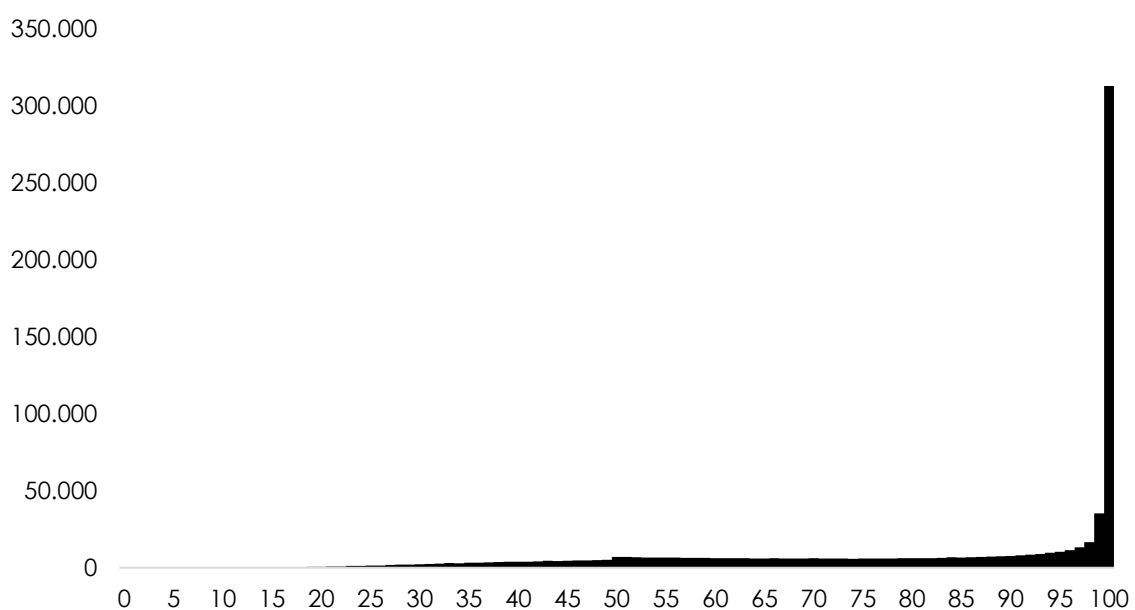
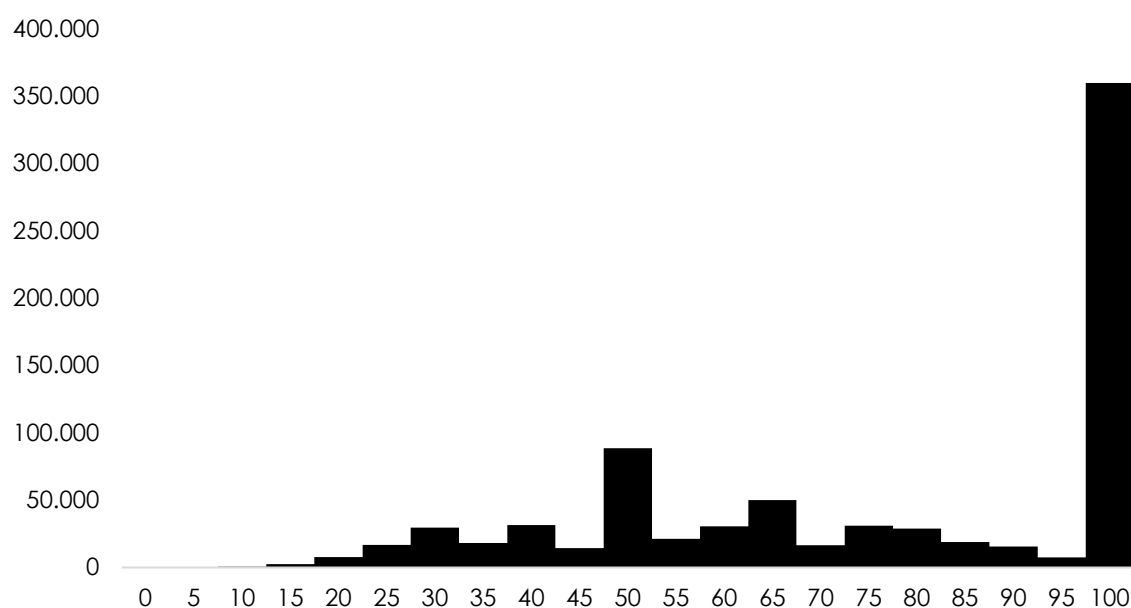
Figure 7. Distribution of the Share of Firm Imports from the Largest Import Source Country

Figure 8. Distribution of the Share of Products Imported from the Dominant Import Partner

To further assess the degree of concentration and potential dependence in firms' sourcing structures, we calculate the Herfindahl-Hirschman Index (HHI) of import concentration. Figure 9 shows that import sourcing at the firm-product level is highly concentrated. The mean HHI remains consistently high, around 0.915 throughout the period, indicating strong reliance on a limited number of supplier countries. Similarly, the share of firm-product combinations with an HHI above 0.4 remains above 96 percent over time. Together, these results suggest that highly concentrated sourcing structures are a persistent feature of firms' import patterns.

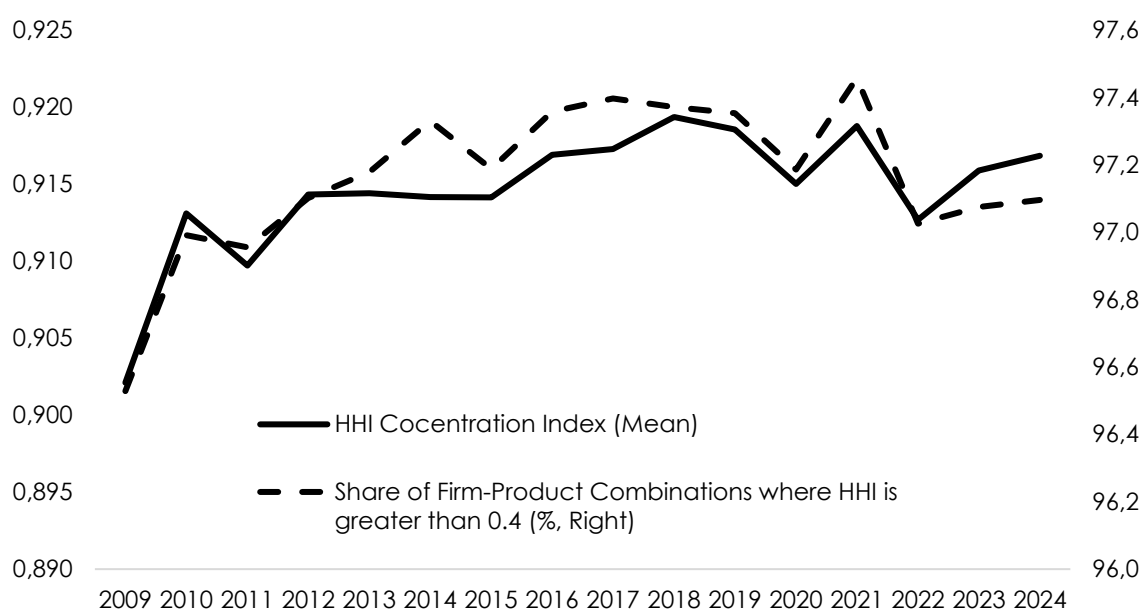
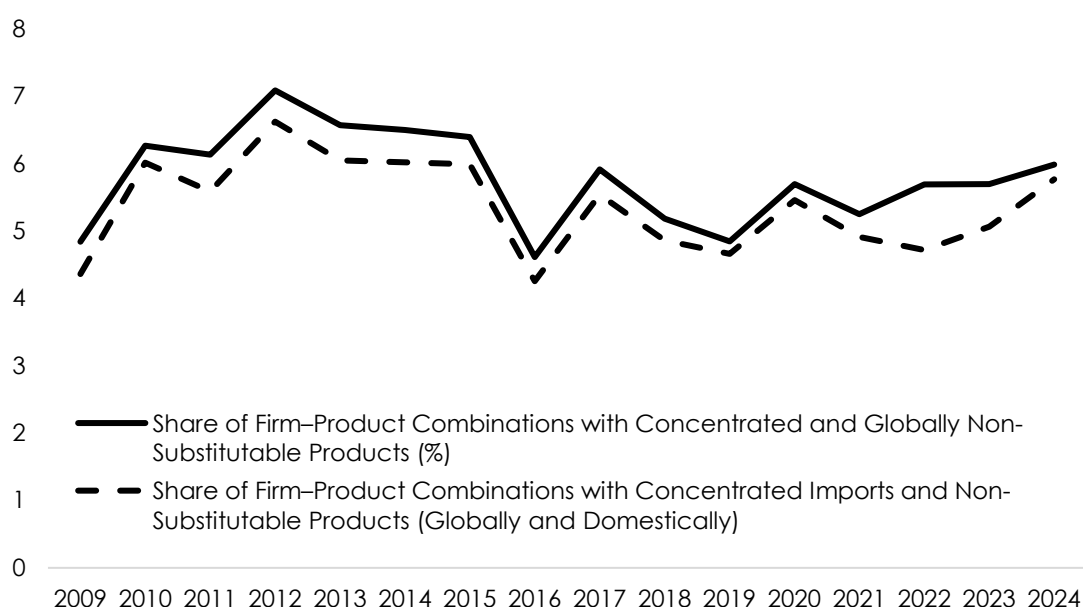
Figure 9. Import Concentration (HHI) and Share of Firm-Product Imports with High Concentration (HHI > 0.4), 2009–2024

Figure 10 reports the share of firm–product import relationships that meet the criteria for concentrated and non-substitutable sourcing. A product is classified as concentrated when the Herfindahl–Hirschman Index (HHI) of imports exceeds 0.4, and non-substitutability is determined using the methodology described above, which evaluates both global supply concentration and the ratio of Dutch imports to exports as an indicator of limited domestic substitution. The results show that around 5–7 percent of firm–product combinations involve imports that are both highly concentrated and difficult to substitute globally. When also accounting for limited domestic substitutability, the share remains slightly lower but follows a similar pattern over time. Overall, the figure suggests that a small but persistent share of imports may expose firms to higher vulnerability due to concentrated and hard-to-substitute supply sources.

Figure 10. Share of Firm–Product Combinations with Concentrated and Non-Substitutable Imports



4. Conclusion and Next Steps

This extended abstract provides a first exploration of import dependencies among Dutch firms using a multidimensional framework that combines firm-level import concentration, global supply concentration, and indicators of domestic substitutability. The preliminary results suggest that firms' sourcing structures are highly concentrated at the firm–product level and that a non-negligible share of imports is both concentrated and difficult to substitute. These patterns highlight potential vulnerabilities in firms' supply chains, particularly when imports rely on a limited number of supplier countries or products with concentrated global supply.

At the same time, this work should be viewed as preliminary. The results presented in this extended abstract mainly provide descriptive evidence on firms' import structures and the prevalence of concentrated sourcing relationships. The ongoing work will extend the analysis by more systematically examining how import dependence relates to firm characteristics

such as firm size, productivity, and firms' upstream and downstream positions in global value chains. In addition, we will further develop the network analysis to assess whether disruptions affecting import-dependent firms could propagate through the Dutch production network and generate broader systemic risks.

Our aim is to complete these additional analyses and refine the empirical results before the conference in June 2026. The final version of the paper will include more detailed analysis, a deeper investigation of firm heterogeneity in import dependence, and a comprehensive assessment of potential contagion effects within the Dutch firm network.

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